BB2 세일 후류 압력장 기반 AUV 상대 위치 추정 모델 개발

Development of a Model for AUV's Relative Position Estimation from the BB2 Sail-Wake Pressure Field

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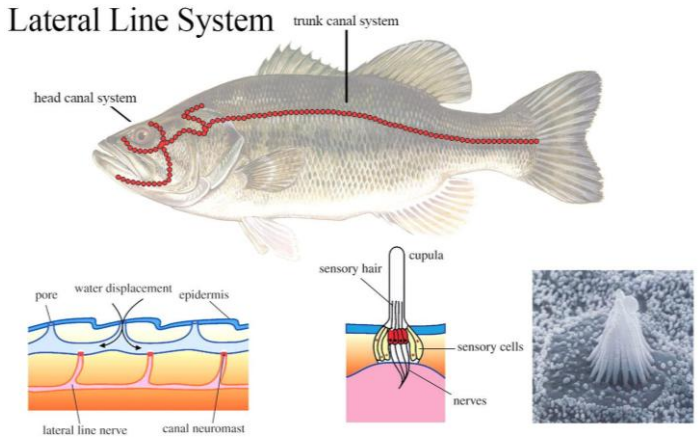
➢ Lateral Line System of Fish

❖ What is the Lateral Line System (LLS) of fish?

- The fish lateral line system (LLS) is a specialized sensory network of mechanoreceptors, called neuromasts, that detects water movements, vibrations, and pressure gradients.
- The LLS enables fish to perceive their environment in dark or murky water and hunt or avoid predators.

❖ Working principle of LLS and application

- Local flow disturbance around the fish produces pressure differences and water displacement near the body surface. These disturbances deform the neuromast structure and are converted into neural signals.
- Inspired by this mechanism, an Artificial Lateral Line System (ALLS) can be constructed using arrays of pressure sensors on the hull surface of autonomous underwater vehicles (AUVs) or robotic fish to detect the motion state.



Lateral Line System



Low-vision : Mexican cavefish



Blind Fish Experiment[1]

[1] Pitcher, T. J., Partridge, B. L., & Wardle, C. S. (1976). A blind fish can school. *Science*, *194*(4268), 963–965. https://doi.org/10.1126/science.982056

➢ AUV Following at a Fixed Offset in a Submarine's Sail Wake

❖ AUV returning after mission completion

- The submarine stays underway at low speed while the AUV approaches.
- We assume the AUV matches the submarine's speed and maintains a fixed offset in the sail wake.

❖ Position estimation with limited sensing

- During recovery, the AUV must estimate its position relative to the submarine.
- Sonar use and other underwater sensing options are constrained

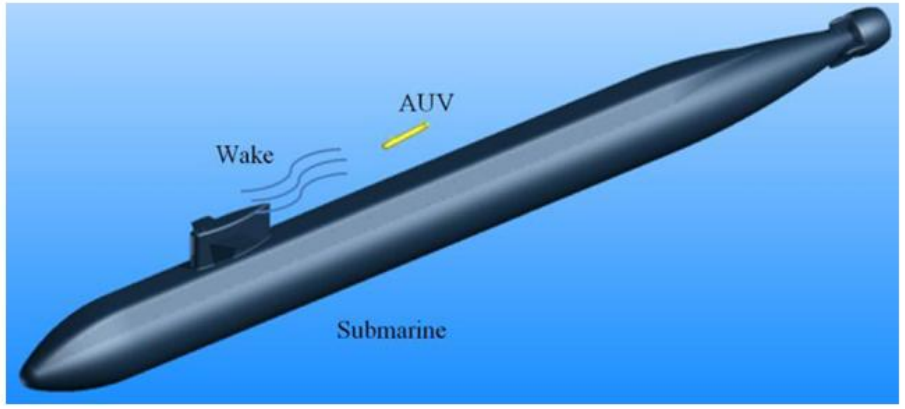
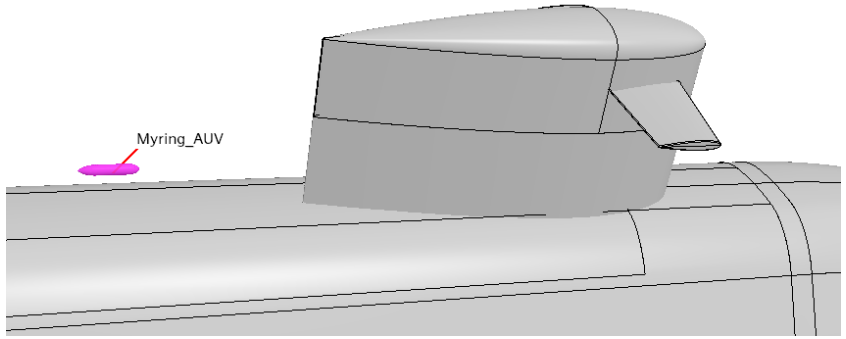


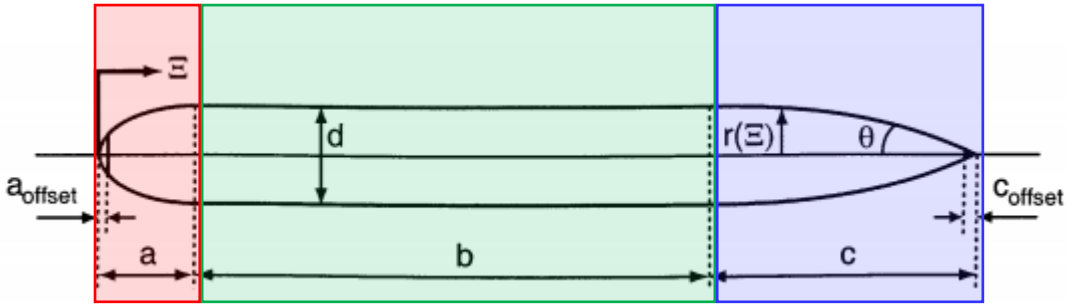
Fig. 1. An AUV moving in the wake of a submarine.



[2] Mai et al. (2024), Establishment of empirical formulae for hydrodynamic derivatives of submarine considering design parameters. International Journal of Naval Architecture and Ocean Engineering, (https://doi.org/10.1016/j.ijnaoe.2023.100537)

➢ **Target Model** : REMUS 100 AUV (w/o control fins)





Nose **Mid Body** **Tail**

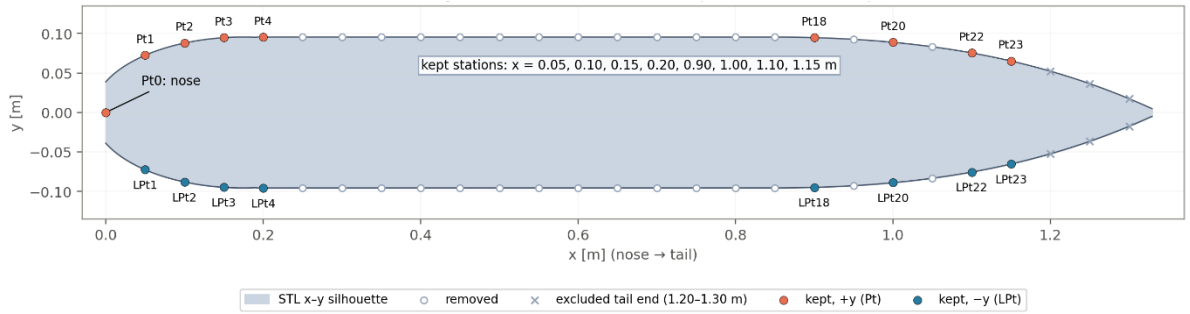
✓ **Myring hull profile equation**

$$r_{nose} = \frac{1}{2}d\left[1-\left(\frac{x+a_{offset}-a}{a}\right)^2\right]^{\frac{1}{n}}$$

$$r_{body} = \frac{1}{2}d$$

$$r_{tail} = \frac{1}{2}d - \left[\frac{3d}{2c^2} - \frac{tan\theta}{c}\right](x-l)^2 + \left[\frac{d}{c^3} - \frac{tan\theta}{c^2}\right](x-l_f)^3$$

Sensor Location on AUV Surface (17 probes)

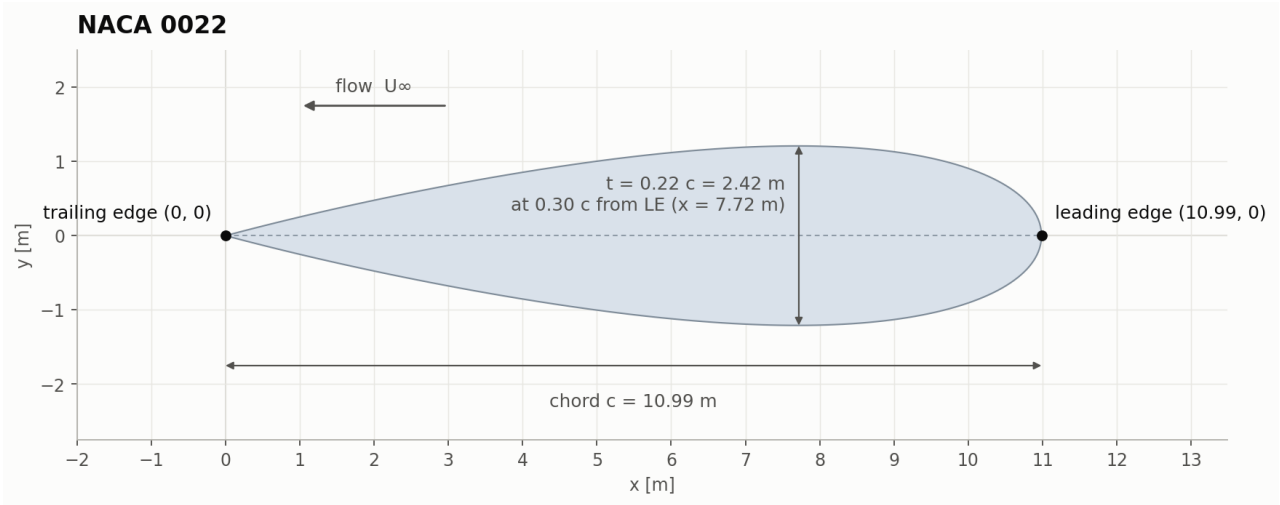


Geometric property of Hull

Properties	value
Nose Length [m]	0.191
Mid Body Length [m]	0.654
Tail Length [m]	0.541
Nose Offset [m]	0.0165
Tail Offset [m]	0.0368
Vehicle Length [m]	1.33
Vehicle Diameter [m]	0.191

➢ **Target Model** : BB2 Submarine's Sail (w/o horizontal sailplanes)

- ❖ The BB2 sail uses a NACA 0022 profile[3].
- ❖ The model omits the horizontal sailplanes.
- ❖ The hull's bottom-wall effect is neglected.
- ❖ The sail chord is c = 10.99 m.



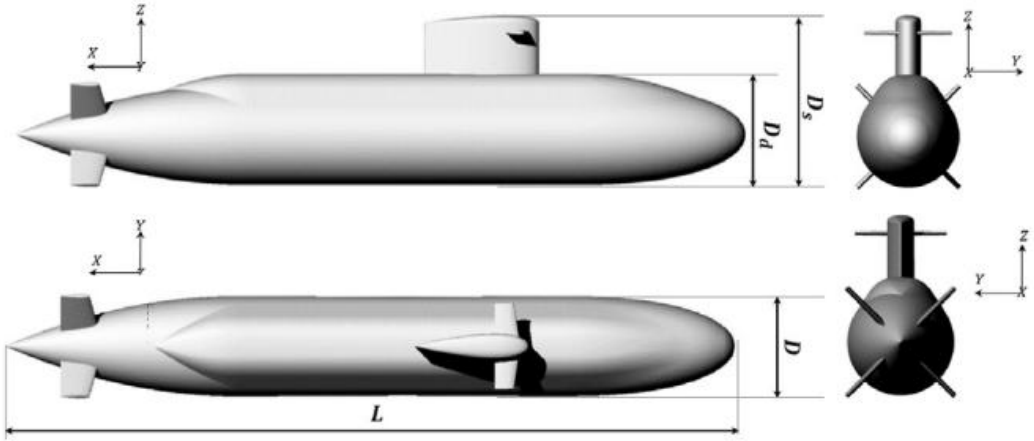


Fig. 1. Geometry of the BB2 submarine.

Properties	value
Length [m]	70.2 [m^3]
Diameter [m]	9.6 [m^3]
Depth to deck [m]	10.6 [m^3]
Depth to sail top [m]	16.2 [m^3]
Displacement volume [m]	4440 [m^3]

[3] Carrica, P. M., Kim, Y., & Martin, J. E. (2019). Near-surface self propulsion of a generic submarine in calm water and waves. *Ocean Engineering*, *183*, 87–105. https://doi.org/10.1016/j.oceaneng.2019.04.082

➢ **Data Generation** : Potential Panel Method & CFD

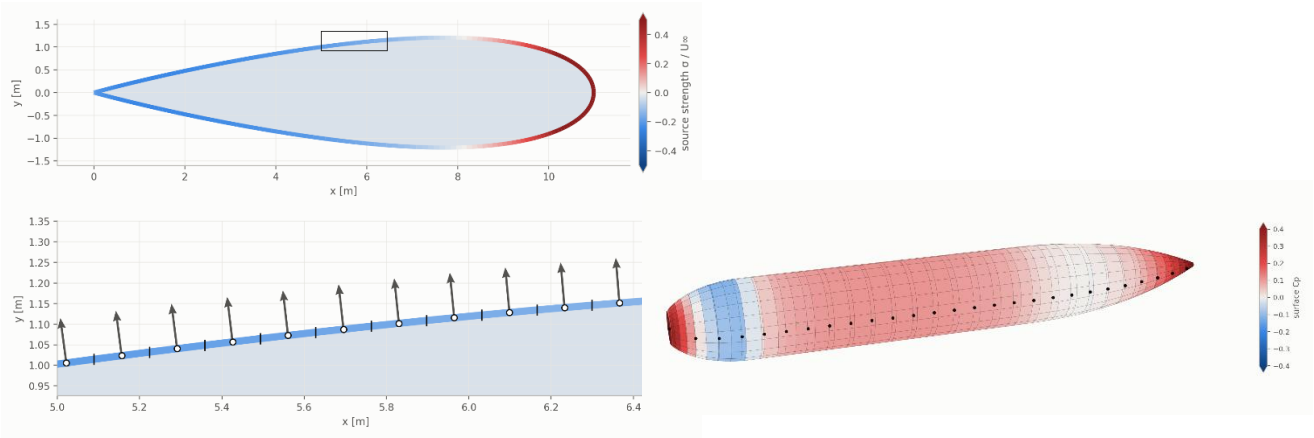
❖ Potential Flow Analysis

- ✓ **Flow assumptions** : Incompressible, inviscid, irrotational
- ✓ **Hull condition** : No penetration
- ✓ **Role** : Rapid dense data sampling where viscous-wake is limited

$$u = \nabla\phi$$

$$\nabla^2\phi = 0 \qquad (n \cdot \nabla\varphi)\Big|_S = 0$$

$$\varphi(x) = U_\infty \cdot x + \sum_{j=1}^{N} \sigma_j G_j(x) \qquad G_j(x) = \int_{S_j} \left(\frac{1}{4\pi|x-\xi|}\right) dS_\xi$$



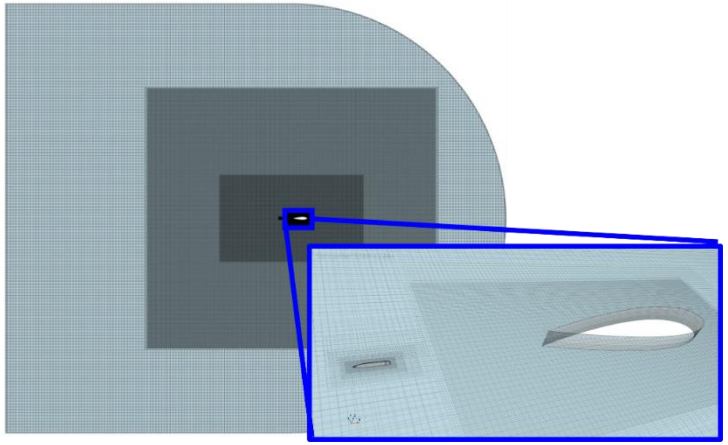
❖ CFD method

- ✓ **Flow assumptions** : Incompressible, viscous RANS
- ✓ **Turbulence model**: $k-\omega$ SST
- ✓ **Role** : Higher-fidelity data sampling including viscosity

$$\nabla \cdot \bar{u} = 0$$

$$\rho\left(\frac{\partial \bar{u}}{\partial t} + (\bar{u} \cdot \nabla)\bar{u}\right) = -\nabla\bar{p} + \nabla \cdot \left[2(\mu + \mu_t)\bar{S} - \frac{2\rho k}{3} I\right]$$

$$\bar{S} = \frac{\nabla\bar{u} + \nabla\bar{u}^T}{2} \qquad \mu_t = \mu_t(k, \omega; SST)$$



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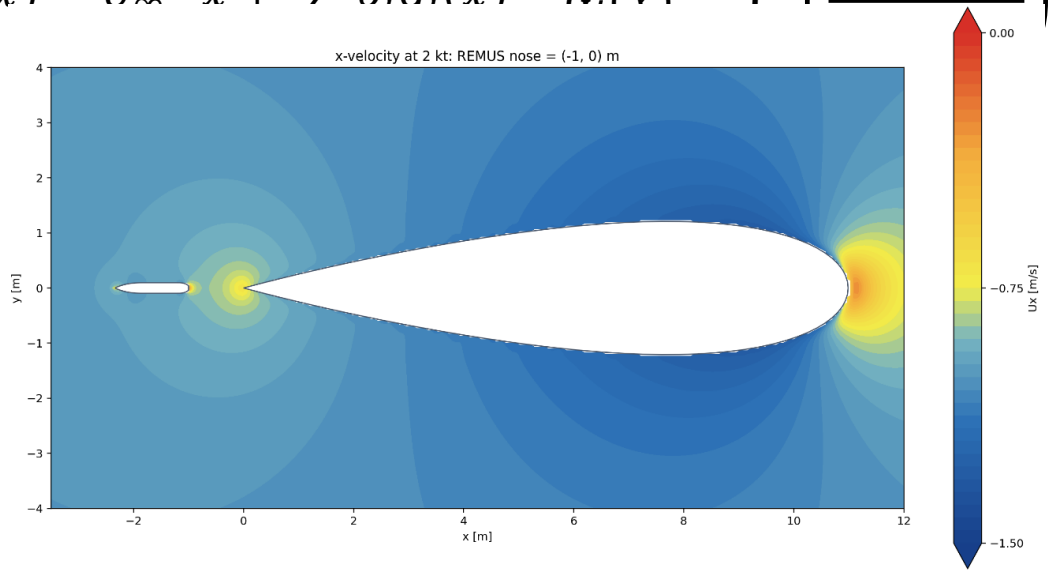
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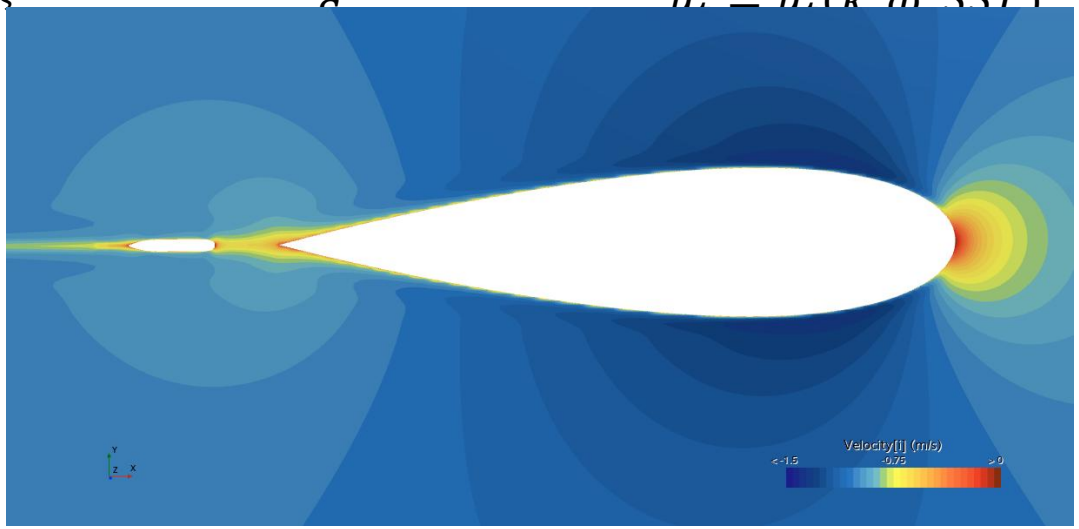
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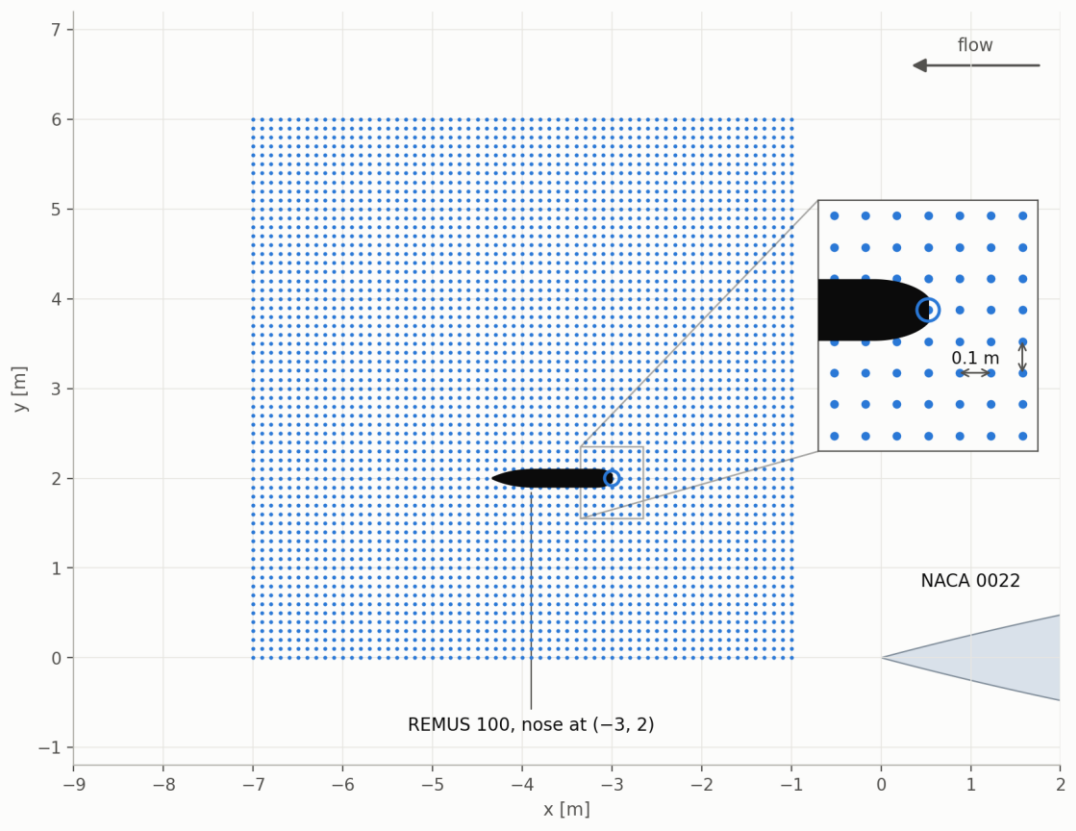
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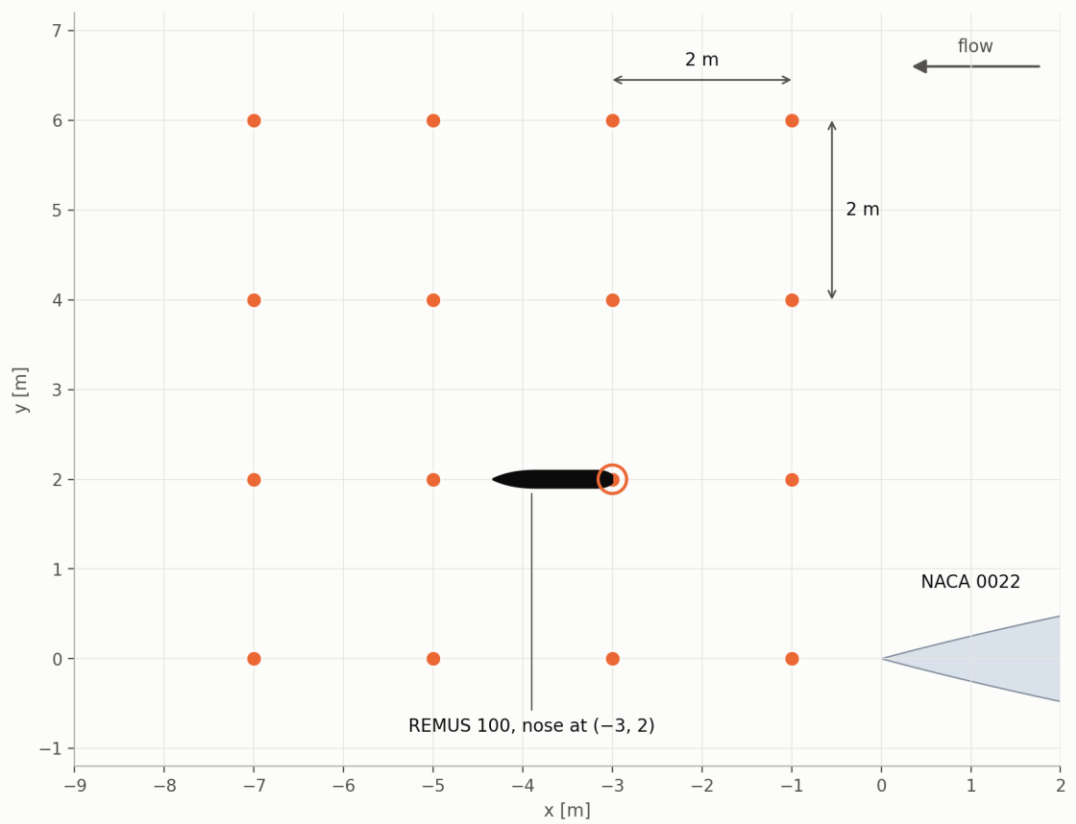
❖ Potential Flow Analysis

- ✓ **Position sampling** : $\Delta s = 0.1\ [m]$
- ✓ **Velocity sampling** : $U_\infty = [1,2,3,4][kts]$



❖ CFD method

- ✓ **Position sampling** : $\Delta s = 2\ [m]$
- ✓ **Velocity sampling** : $U_\infty = [2,3][kts]$



➢ Estimation Models

❖Notation & Pressure Model

$$\mathbf{p} = [\,p_1,\ \ldots,\ p_n\,]^\top$$

$$q(U_\infty) = \frac{1}{2}\,\rho\,U_\infty^2$$

$$p_k^{\mathrm{pot}}(\mathbf{x}, U_\infty) = C_{p,k}^{\mathrm{pot}}(\mathbf{x})\,q(U_\infty)$$

$$r_k = p_k^{\mathrm{CFD}} - p_k^{\mathrm{pot}}(\mathbf{x}, U_\infty)$$

$$\mathbf{z}(\mathbf{p}) = (\mathbf{p} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$$

$$\mathbf{x} = (x,\ y), \qquad \boldsymbol{\theta} = [\,x,\ y,\ U_\infty\,]^\top$$

$$\mathcal{D} = \left\{\left(\mathbf{z}(\mathbf{p}_i^{\mathrm{pot}}),\ \boldsymbol{\theta}_i\right)\right\}_{i=1}^{M}, \qquad \mathbf{p}_i^{\mathrm{pot}} = \mathbf{C}_p^{\mathrm{pot}}(\mathbf{x}_i)\,q(U_{\infty,i}), \quad U_{\infty,i} \in \{2,\ 3\}\ \mathrm{kt}$$

$$\mathbf{z}_* = \mathbf{z}(\mathbf{p}_*), \qquad \mathbf{p}_* \in \{\mathbf{p}^{\mathrm{CFD}},\ \mathbf{p}^{\mathrm{corr}}\}$$

❖Weighted KNN

$$\hat{\boldsymbol{\theta}}(\mathbf{z}_*) = \frac{\sum_{i \in \mathcal{N}_5(\mathbf{z}_*)} w_i\,\boldsymbol{\theta}_i}{\sum_{i \in \mathcal{N}_5(\mathbf{z}_*)} w_i}$$

$$w_i = \frac{1}{\|\mathbf{z}_* - \mathbf{z}_i\|^2}$$

$$[\,\hat{x},\ \hat{y},\ \hat{U}_\infty\,]^\top = \hat{\boldsymbol{\theta}}(\mathbf{z}_*)$$

❖Gaussian Process regression (ARD kernel)

$$k(\mathbf{z}, \mathbf{z}') = \exp\!\Big(-\frac{1}{2}\sum_{j=1}^{n}\frac{(z_j - z_j')^2}{\ell_j^2}\Big)$$

$$\hat{\theta}_m(\mathbf{z}_*) = \bar{\theta}_m + \sigma_{\theta,m}\,\mathbf{k}_*^\top(\mathbf{K} + \sigma_\varepsilon^2\mathbf{I})^{-1}\,\tilde{\boldsymbol{\theta}}_m, \qquad m \in \{x,\ y,\ U_\infty\}$$

$$\tilde{\theta}_{i,m} = \frac{\theta_{i,m} - \bar{\theta}_m}{\sigma_{\theta,m}}, \qquad K_{ij} = k(\mathbf{z}_i, \mathbf{z}_j), \quad (\mathbf{k}_*)_i = k(\mathbf{z}_*, \mathbf{z}_i)$$

$$\ell_j = \min\left\{20,\ \max\left\{0.25,\ \frac{1}{\max_m|\mathrm{corr}(z_j, \theta_m)|}\right\}\right\}$$

$$[\,\hat{x},\ \hat{y},\ \hat{U}_\infty\,]^\top = [\,\hat{\theta}_x,\ \hat{\theta}_y,\ \hat{\theta}_{U_\infty}\,]^\top$$

❖Probabilistic inverse estimation

$$\hat{\mathbf{p}}(\mathbf{x}, U_\infty) = \mathbf{C}_p^{\mathrm{pot}}(\mathbf{x})\,q(U_\infty)$$

$$L(\mathbf{x}, U_\infty) = \frac{1}{2}\sum_{k=1}^{n}\frac{\left(p_k^{\mathrm{corr}} - \hat{p}_k(\mathbf{x}, U_\infty)\right)^2}{q(U_\infty)^2\,s_k^2} + \sum_{k=1}^{n}\ln\!\left(q(U_\infty)\,s_k\right)$$

$$(\hat{x},\ \hat{y},\ \hat{U}_\infty) = \underset{(x,y)\in\mathcal{G},\ U_\infty\in\mathcal{U}}{\arg\min}\ L\big((x, y), U_\infty\big)$$

$$\mathcal{G} = \{-7,\ -6.9,\ \ldots,\ -1\} \times \{0,\ 0.1,\ \ldots,\ 6\}\ \mathrm{m}, \qquad \mathcal{U} = \{1.5,\ 1.55,\ \ldots,\ 3.5\}\ \mathrm{kt}$$

$$s_k^2 = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\hat{r}_k^{(-j_i)}(\mathbf{p}_i) - r_{ik}}{q(U_{\infty,i})}\right)^2$$

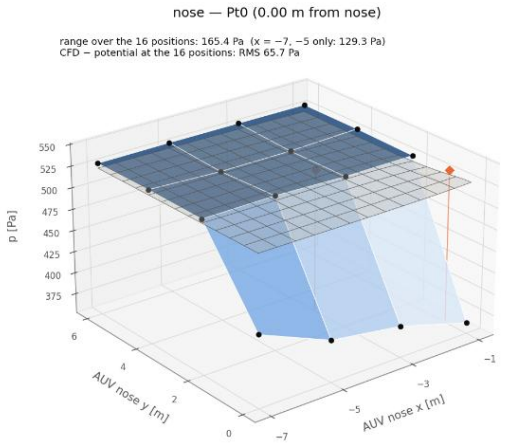
➢ Correction Model

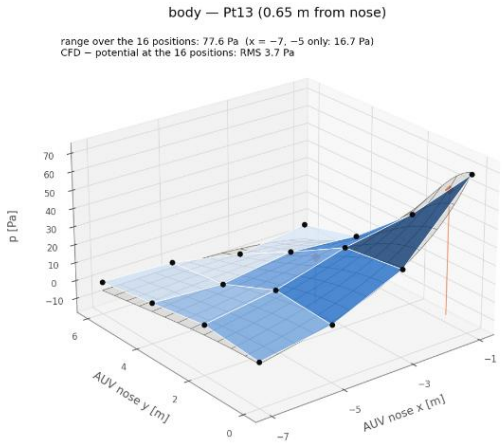
❖Ridge

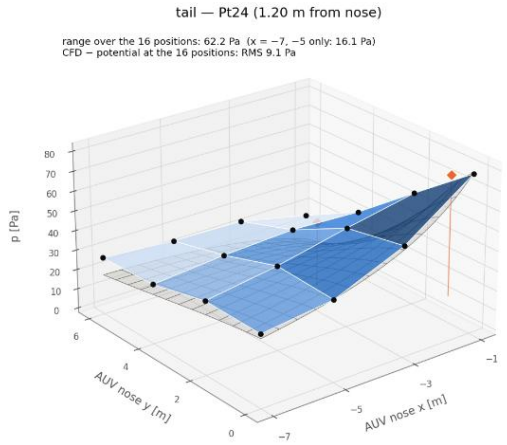
$$\hat{\mathbf{r}}(\mathbf{p}) = \mathbf{m} + \mathbf{W}^{\top}\mathbf{z}(\mathbf{p})$$

$$\mathbf{W} = (\mathbf{Z}^{\top}\mathbf{Z} + \alpha\mathbf{I})^{-1}\,\mathbf{Z}^{\top}(\mathbf{R} - \mathbf{1}\mathbf{m}^{\top})$$

$$\mathbf{p}^{\mathrm{corr}} = \mathbf{p}^{\mathrm{CFD}} - \hat{\mathbf{r}}(\mathbf{p}^{\mathrm{CFD}})$$





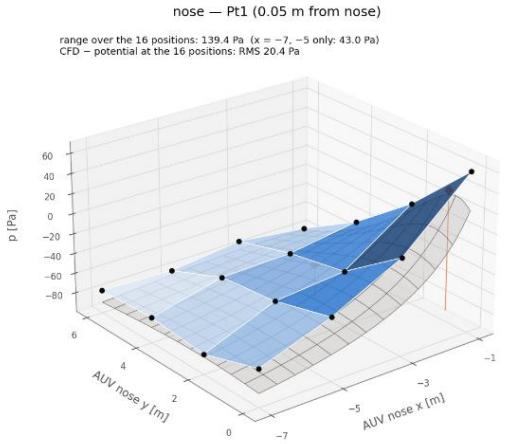


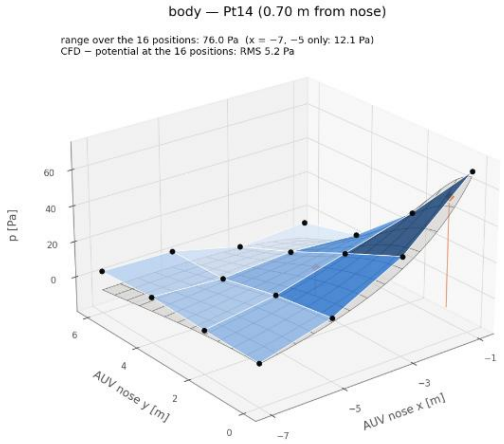
❖Gaussian Process

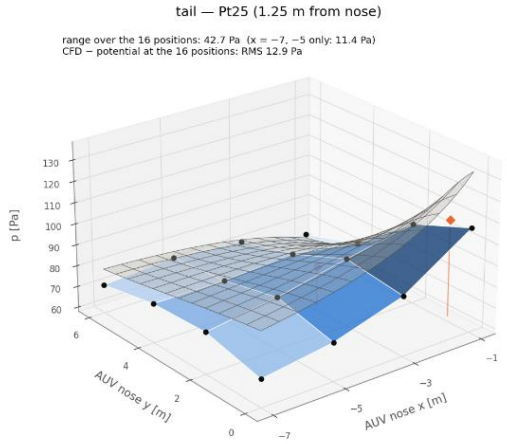
$$\mathbf{s}(\mathbf{p}) = \mathbf{V}_m^{\top}\,\mathbf{z}(\mathbf{p})$$

$$\kappa(\mathbf{s}, \mathbf{s}') = \exp\Big(-\frac{\|\mathbf{s} - \mathbf{s}'\|^2}{2\ell^2}\Big)$$

$$\hat{\mathbf{r}}(\mathbf{p}) = \mathbf{m} + (\mathbf{R} - \mathbf{1}\mathbf{m}^{\top})^{\top}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\,\boldsymbol{\kappa}_*$$



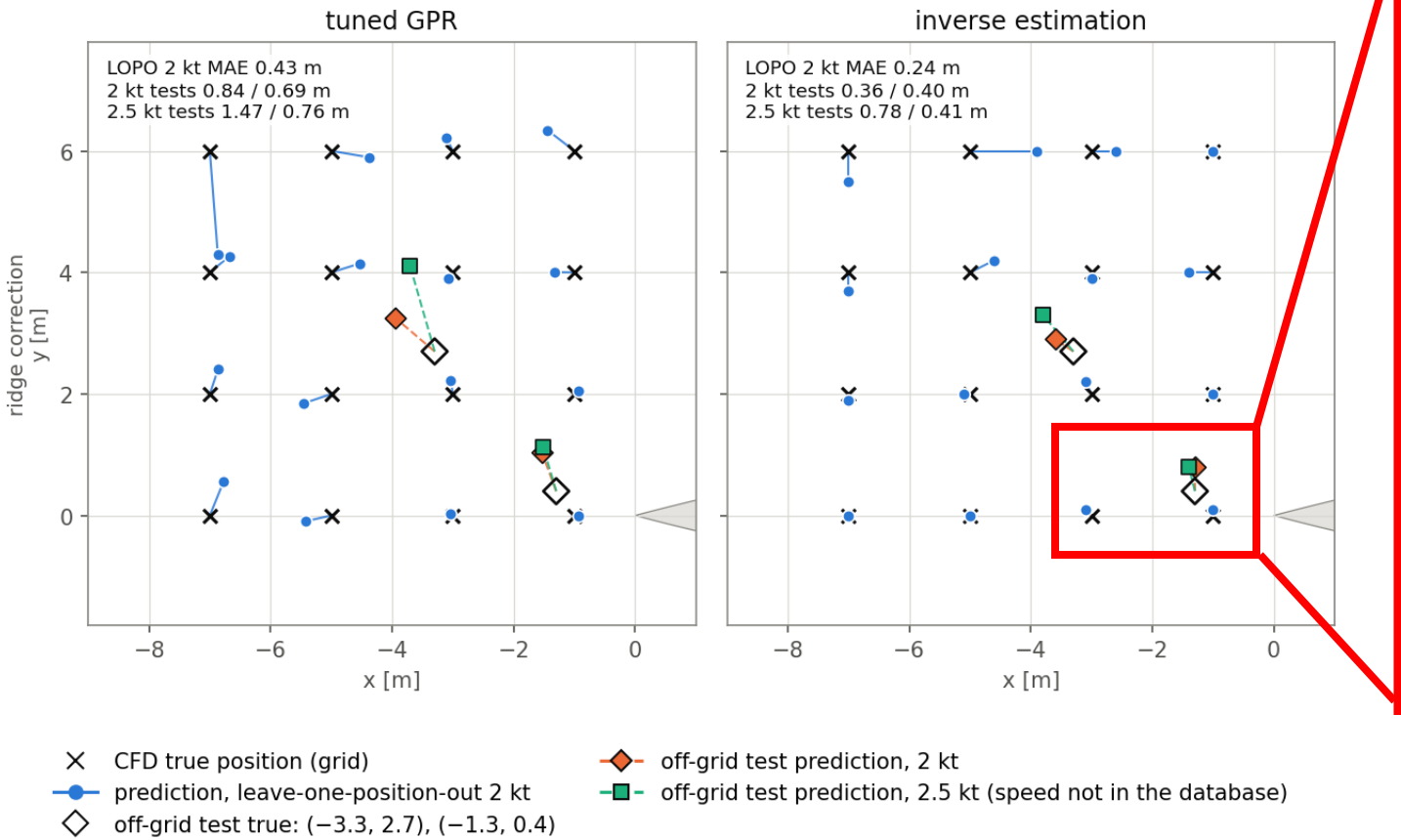


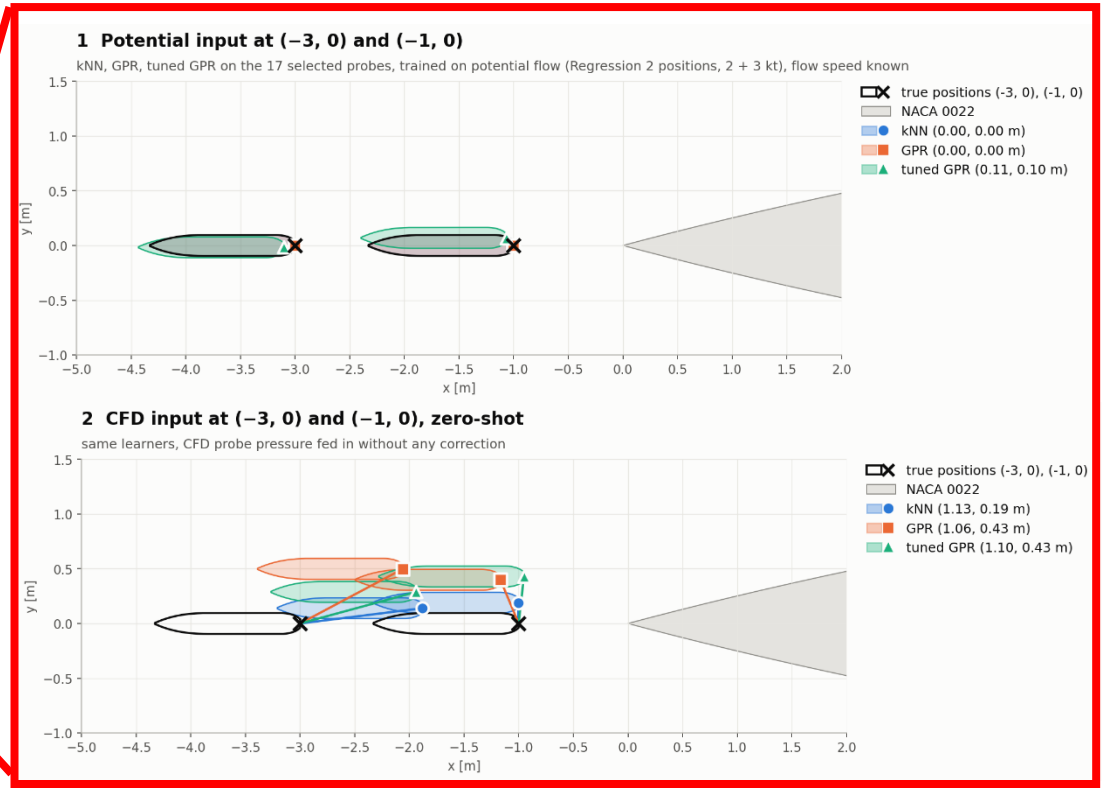


Results

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➢ Position Estimation Results





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